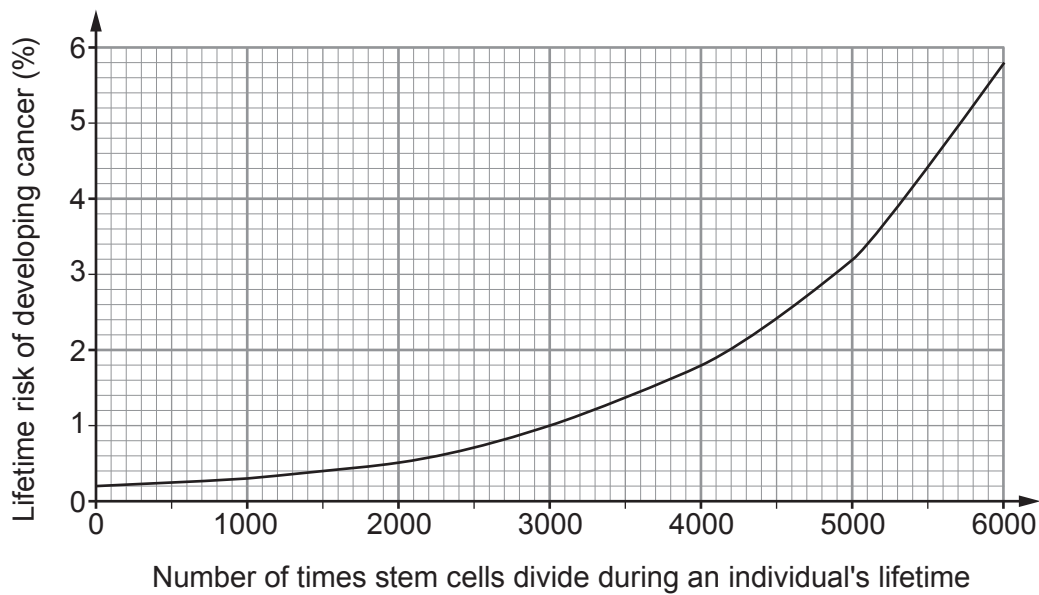


9. In humans, the number of times a stem cell undergoes cell division varies. For example, some stem cells in the skin will divide up to 200 times during an individual's lifetime whilst other stem cells found in the large intestine may divide up to 5800 times. The lifetime risk of developing a cancer is different for each type of tissue in the human body. For example, the lifetime risk of developing skin cancer is 2%. The lifetime risk of developing a cancer of the liver is 0.71%. Scientists have suggested that some of the variation in cancer risk can be explained by the number of times stem cells divide in each tissue during an individual's lifetime, as shown in **Graph 9.1**.

Graph 9.1



- (a) (i) State what is meant by a stem cell. [2]

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- (ii) State the name of the type of cell division responsible for growth and explain how it can lead to cancer. [2]

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Examiner
only

- (b) (i) Use all the information given to identify the percentage lifetime risk of developing cancer of the large intestine. [1]

Lifetime risk = %

- (ii) Suggest **one** reason why the incidence of cancer is rare amongst young people, but tends to increase with age. [1]

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- (c) Other than environmental factors and the number of times a cell divides, name **one** other factor which could be responsible for the variation in the risk of cancers in different tissues. [1]

.....

7

